

Supplementary Materials

A. NOTATION TABLE

Notation	Description
R	Multivariate time series
w	Length of the protected window under w -event DP
ℓ	Segment length
γ	Stride for time series segmentation
$\mathcal{S}$	Set of time series segments
n_s	Number of w -span segments
λ	Window scaling factor in Poisson spaced sampling
ξ	Window sampling probability in Poisson spaced sampling
ζ	Window size under Poisson spaced sampling

The table above summarizes the key notation used in this paper.

B. BACKGROUND OF THE STRUCTURED SAMPLING

We first provide the background on the univariate time series setting.

DEFINITION (UNIVARIATE TIME SERIES SETTING [38]). A time series dataset in the univariate setting is a collection of N univariate time series, denoted by $\mathcal{D} = \{X^{(1)}, X^{(2)}, \dots, X^{(N)}\}$, where $X^{(i)} \in \mathbb{R}^{L_{ts}}$ denotes the i -th univariate time series of length L_{ts} .

Under the univariate setting, the time series within a dataset are considered independent of one another. In the univariate setting with event-level DP, structured sampling adopts a bi-level sampling procedure: top-level sampling first selects a subset of time series, and bottom-level sampling then draws one piece of subsequence from each selected series to form a batch [38], as illustrated in Fig. 1. This design bounds the sensitivity by limiting the contribution of each time series to a single sample. For this setting, Schuchardt et al. [38] derived tight event-level privacy guarantees through PLD-based accounting. When multiple subsequences are drawn from the same selected time series, the framework considers different bottom-level sampling schemes, including Poisson sampling and sampling with replacement [38]. In our multivariate setting, the dataset consists of a single high-dimensional time series with correlated variables, making the sensitivity-control mechanism of structured sampling inapplicable. For the experimental comparison under w -event DP, we implement structured sampling following the setting considered in the original paper for the case where multiple training samples are drawn from a single time series [38]. The paper employs sampling with replacement, but its privacy accounting upper bound is not guaranteed to be tight when multiple training samples are drawn from the same time series [38]. Moreover, we provide a summary of the existing sampling mechanisms in Table 1 for clarity.

C. COMPLETE PROOF

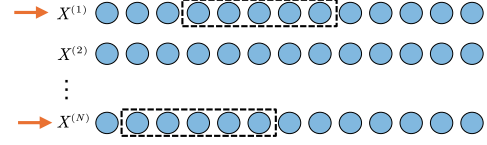


Figure 1: The structured sampling first randomly selects a subset of time series and then draws one training sample from each selected series to form a batch.

In RDP accounting, the privacy loss is given by

$$\tau = \frac{i}{\alpha - 1} \log \left(\sum_{k=0}^m \binom{m}{k} (1-q)^{m-k} q^k \exp \left(\frac{(\alpha-1)ak^2}{2\sigma^2} \right) \right),$$

where m denotes the maximum number of sampled w -span segments in each batch. Notably, $m = n_s$ under Poisson sampling. Under spaced sampling, each w -span segment is assigned to exactly one window. This restriction limits the number of sampled w -span segments per batch, thereby reducing the accumulated privacy loss. However, the value of m remains dependent on the specific segment allocation. For example, Fig. 2 illustrates that all seven w -span segments can be allocated across either two or three windows, resulting in either $m = 2$ or $m = 3$.

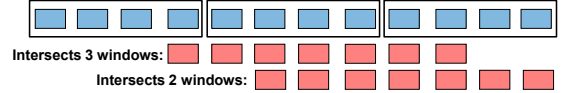


Figure 2: Seven w -span segments are allocated across two or three windows.

To characterize the maximum possible privacy loss, we seek the allocation of w -span segments that maximizes the privacy loss. We first analyze the cases $n_s < t$ and $t \leq n_s < 2t$, and then extend the argument by induction to the general case $n_s \geq 2t$.

1 Proof of Case 1

CASE 1. Let t denote the window size. To derive an upper bound on the privacy loss, we start from the simple case $n_s < t$. In this case, the w -span segments can span at most two windows, as illustrated in Fig. 3.



Figure 3: w -span segments allocated across one or two windows when $n_s < t$.

Intuitively, a w -span segment incurs higher privacy loss when it is assigned to a window of size t than to a window of size $t + 1$. Therefore, we use the window size t to derive an upper bound on the privacy loss. The conclusion is described below.

LEMMA 4.1. Given a batch size K and a set of segments $\mathcal{S}$ obtained by partitioning the time series chronologically, let the window size $t = \lfloor |\mathcal{S}|/K \rfloor$ and suppose the number of w -span segments $n_s < t$. The maximum privacy loss is attained when the w -span segments are evenly allocated between two windows.

Table 1: Existing sampling mechanisms and their limitations under w -event DP for multivariate time series learning.

Method	Original Scenarios	Configuration	Multivariate Setting Applicability	Limitations of Adaptation to Multivariate Setting under w -event DP
Poisson sampling [4]	Multiple univariate time series & event-level DP	Independent inclusion with fixed probability for each segment	Directly applicable and therefore used as the primary baseline	(1) Selecting multiple segments from any w -length subsequence reduces privacy amplification
Structured sampling [38]	Multiple univariate time series & event-level DP	Hierarchical sampling with top-level and bottom-level sampling modules	Retaining only the bottom-level sampler loses the benefits of hierarchical sampling	(2) Ignoring concept drift leads to imbalanced batches and degraded model utility

PROOF. We analyze the privacy loss in two steps. **(1) Formulate the privacy loss.** Since the number of w -span segments n_s is smaller than the window size $t = \lfloor |S|/K \rfloor$, the w -span segments are contained in at most two windows. Let x denote the number of w -span segments included in one window, the upper bound of privacy loss can be denoted as

$$\begin{aligned} \tau &= \frac{1}{\alpha-1} \log\left(\frac{t-x}{t} \cdot \frac{t-(n_s-x)}{t} \exp((\alpha-1)D_\alpha^0)\right) \\ &\quad + \left(\frac{x}{t} \cdot \frac{t-(n_s-x)}{t} + \frac{t-x}{t} \cdot \frac{n_s-x}{t}\right) \exp((\alpha-1)D_\alpha^1) \\ &\quad + \frac{x}{t} \cdot \frac{n_s-x}{t} \exp((\alpha-1)D_\alpha^2)). \end{aligned}$$

where D_α^m ($m = 0, 1, 2$) represents the Rényi divergence when m w -span segments are included. Since the sensitivity of the clipped gradient is $2C$ per sample, we add Gaussian noise with distribution $\mathcal{N}(0, 4C^2\sigma^2\mathbf{I})$. Let m be the number of selected samples from the w -span segments, B and B' be the batches from w -neighboring time series, and let f be the training model. The Rényi divergence can be bounded as

$$\begin{aligned} D_\alpha(\mathcal{N}(f(B), 4C^2\sigma^2\mathbf{I}) \| \mathcal{N}(f(B'), 4C^2\sigma^2\mathbf{I})) \\ &= D_\alpha(\mathcal{N}(f(B) - f(B'), 4C^2\sigma^2\mathbf{I}) \| \mathcal{N}(0, 4C^2\sigma^2\mathbf{I})) \\ &\leq \sup_{\|v\|_2 \leq 2mC} D_\alpha(\mathcal{N}(v, 4C^2\sigma^2\mathbf{I}) \| \mathcal{N}(0, 4C^2\sigma^2\mathbf{I})) \\ &= \sup_{\|v\|_2 \leq 2mC} \frac{\alpha \|v\|_2}{2 \cdot 4C^2\sigma^2} = \frac{\alpha m^2}{2\sigma^2}. \end{aligned}$$

Therefore, the upper bound of privacy loss above can be derived as

$$\begin{aligned} \tau &= \frac{1}{\alpha-1} \log\left(\frac{t-x}{t} \cdot \frac{t-(n_s-x)}{t}\right) \\ &\quad + \left(\frac{x}{t} \cdot \frac{t-(n_s-x)}{t} + \frac{t-x}{t} \cdot \frac{n_s-x}{t}\right) \exp\left(\frac{\alpha(\alpha-1)}{2\sigma^2}\right) \\ &\quad + \frac{x}{t} \cdot \frac{n_s-x}{t} \exp\left(\frac{4\alpha(\alpha-1)}{2\sigma^2}\right) \\ &= \frac{1}{\alpha-1} \log(\text{Const} + \frac{1-2e^c + e^{4c}}{t^2} x(n_s-x)). \end{aligned} \quad (2)$$

where Const denotes the constant term and $c = \frac{\alpha(\alpha-1)}{2\sigma^2}$. **(2) Maximize the privacy loss for fixed α .** Let $a = e^c > 1$ and define $h(a) = a^4 - 2a + 1$. Then, we have $h(a) > h(1) = 0$ for all

$a > 1$. Consequently, the maximized upper bound of privacy loss $\tau_{\max}$ is attained at $x = \lceil \frac{n_s}{2} \rceil$. $\square$

Compare the privacy costs across different window sizes.

Intuitively, a w -span segment incurs higher privacy loss when it is assigned to a window of size t than when it is assigned to a window of size $t+1$, since the sampling rate of a w -span segment is smaller in a window of size $t+1$. Let $A = e^c - 1$, $B = e^{4c} - 2e^c + 1$, and $c = \frac{\alpha(\alpha-1)}{2\sigma^2}$. Let $F_{t,t}$ denote the logarithmic term of Eq. (2), $F_{t,t} = 1 + \frac{An_s}{t} + \frac{B}{t^2}x(n_s-x)$. The privacy loss $\tau_{t,t+1}$ of the two windows with size t and $t+1$ is

$$\begin{aligned} \tau_{t,t+1} &= \frac{1}{\alpha-1} \log\left(\frac{t-x}{t} \cdot \frac{t+1-(n_s-x)}{t+1}\right) \\ &\quad + \left(\frac{x}{t} \cdot \frac{t+1-(n_s-x)}{t+1} + \frac{t-x}{t} \cdot \frac{n_s-x}{t+1}\right) \exp\left(\frac{\alpha(\alpha-1)}{2\sigma^2}\right) \\ &\quad + \frac{x}{t} \cdot \frac{n_s-x}{t+1} \exp\left(\frac{4\alpha(\alpha-1)}{2\sigma^2}\right). \end{aligned}$$

Let $F_{t,t+1}$ denote the logarithmic term, $F_{t,t+1} = 1 + A(\frac{x}{t} + \frac{n_s-x}{t+1}) + \frac{B}{t(t+1)}x(n_s-x)$. Since $F_{t,t} - F_{t,t+1} = \frac{n_s-x}{t(t+1)}A + \frac{x(n_s-x)}{t^2(t+1)}B \geq 0$, it follows that $F_{t,t} \geq F_{t,t+1}$. Therefore, the maximum privacy loss is attained when the w -span segments are allocated to windows of size t .

2 Proof of Case 2

Case 2: $t \leq n_s < 2t$. We next consider the case $t \leq n_s < 2t$. In this regime, the w -span segments can occupy two or three windows as shown in Fig. 4.



Figure 4: w -span segments allocated across one or two windows when $n_s < t$.

By Lemma 4.2, when the w -span segments are contained in two windows, the privacy loss is maximized when they are split evenly between the two windows. When they are contained in three windows, the privacy loss is maximized when one window is fully occupied by w -span segments and the remaining w -span segments are split evenly between the two windows adjacent to the fully occupied window. The corresponding maximum privacy loss is analyzed below.

LEMMA 4.2. *Given a batch size K and a set of segments $\mathcal{S}$. Let $t = \lfloor |S|/K \rfloor$ and suppose the number of w -span segments satisfies*

$t < n_s < 2t$. Define $c = \frac{\alpha(\alpha-1)}{2\sigma^2}$, where α is the order of Rényi divergence and σ is the noise multiplier of the Gaussian mechanism. Let $A = 1 - 3e^c + 3e^{4c} - e^{9c}$, $B = (2\rho - 1) + (2 - 4\rho)e^c + (2\rho - 1)e^{4c}$, $C = \left(\rho - \frac{1}{2}\right)^2 - \left(2\rho^2 + \frac{1}{2}\right)e^c + \left(\rho + \frac{1}{2}\right)^2 e^{4c}$, where $\rho \in \{0, \frac{1}{2t}, -\frac{1}{2t}\}$ is the discretization error induced by the ceiling operation. Define

$$b_0 = \begin{cases} b^*, & \text{if } B^2 - 4AC \geq 0 \text{ and } b^* = \frac{-B - \sqrt{B^2 - 4AC}}{2A} \in [0, \frac{1}{2}), \\ 0, & \text{otherwise.} \end{cases}$$

Then the privacy loss attains its maximum value:

- 1) If $n_s - t \leq 2\lceil b_0 t \rceil - 2$, the w -span segments are evenly split between two windows.
- 2) Otherwise, one window is first filled entirely with w -span segments and the remaining w -span segments are then equally split between the two boundary windows adjacent to the fully occupied window.

PROOF. We analyze the privacy loss case by case. **(1) Derive the privacy loss τ_2 when the w -span segments are distributed across two windows.** Let n_s be the number of w -span segments, K be the batch size, $t = \lfloor |S|/K \rfloor$, and $D_\alpha^m = \frac{\alpha m^2}{2\sigma^2}$ ($m = 0, 1, 2, 3$) represent the Rényi divergence when m w -span segments are included in one batch. Let τ_2 denote the privacy loss when the w -span segments are contained in two windows,

$$\tau_2 = \frac{1}{\alpha - 1} \log\left(\frac{(t-x)^2}{t^2} \exp((\alpha-1)D_\alpha^0) + 2 \cdot \frac{x}{t} \cdot \frac{t-x}{t} \exp((\alpha-1)D_\alpha^1) + \frac{x^2}{t^2} \exp((\alpha-1)D_\alpha^2)\right),$$

where $x = \lceil \frac{n_s}{2} \rceil$, and we have $\frac{x}{t} \in [0.5, 1)$. **(2) Derive the privacy loss τ_3 when the w -span segments are distributed across three windows.** Similarly, the privacy loss τ_3 when the w -span segments are contained in three windows can be denoted as

$$\tau_3 = \frac{1}{\alpha - 1} \log\left(\frac{(t-y)^2}{t^2} \exp((\alpha-1)D_\alpha^1) + 2 \cdot \frac{y}{t} \cdot \frac{t-y}{t} \exp((\alpha-1)D_\alpha^2) + \frac{y^2}{t^2} \exp((\alpha-1)D_\alpha^3)\right),$$

where $y = \lceil \frac{n_s-t}{2} \rceil$, and hence $\frac{y}{t} \in [0, 0.5)$. **(3) Choose the maximum privacy loss between τ_2 and τ_3 .** Since $\log(\cdot)$ is a monotonically increasing function, it suffices to compare τ_2 and τ_3 by comparing their logarithmic terms. Let $b = \frac{y}{t} \in [0, 0.5)$ and denote $a = \frac{x}{t}$. For ease of analysis, we write $a = \frac{1}{2} + b + \rho$, where ρ denotes the discretization error induced by the ceiling operation and $\rho \in \{0, \frac{1}{2t}, -\frac{1}{2t}\}$.

The following explains $\rho \in \{0, \frac{1}{2t}, -\frac{1}{2t}\}$. Let

$$x = \left\lceil \frac{n_s}{2} \right\rceil, \quad y = \left\lceil \frac{n_s-t}{2} \right\rceil,$$

and define $a = \frac{x}{t}$, $b = \frac{y}{t}$, we can write

$$a = \frac{x}{t} = \frac{1}{2} + \frac{y}{t} + \rho = \frac{1}{2} + b + \rho,$$

where

$$\rho = \frac{x-y-\frac{t}{2}}{t}.$$

If $t = 2k$ is even, then $t/2 = k \in \mathbb{Z}$, and hence

$$x = \left\lceil \frac{n_s-t}{2} + k \right\rceil = \left\lceil \frac{n_s-t}{2} \right\rceil + k = y + k.$$

Therefore,

$$\rho = \frac{x-y-\frac{t}{2}}{t} = \frac{(y+k)-y-k}{t} = 0.$$

If $t = 2k + 1$ is odd, write

$$u = \frac{n_s-t}{2} = m + r, \quad m \in \mathbb{Z}, r \in [0, 1).$$

Then

$$y = \lceil u \rceil, \quad x = \left\lceil u + k + \frac{1}{2} \right\rceil = k + \left\lceil u + \frac{1}{2} \right\rceil.$$

Thus

$$x - y - \frac{t}{2} = \left\lceil u + \frac{1}{2} \right\rceil + k - \lceil u \rceil - \frac{1}{2} - k = \left\lceil u + \frac{1}{2} \right\rceil - \lceil u \rceil - \frac{1}{2}.$$

Now, according to the fractional part r of u ,

$$\left\lceil u + \frac{1}{2} \right\rceil - \lceil u \rceil - \frac{1}{2} = \left\lceil m + r + \frac{1}{2} \right\rceil - \lceil m + r \rceil - \frac{1}{2} = \begin{cases} \frac{1}{2}, & r = 0 \text{ or } \frac{1}{2} < r < 1, \\ -\frac{1}{2}, & 0 < r \leq \frac{1}{2}. \end{cases}$$

Hence,

$$\rho = \frac{x-y-\frac{t}{2}}{t} \in \left\{ \frac{1}{2t}, -\frac{1}{2t} \right\}.$$

Combining the even and odd cases yields

$$\rho \in \left\{ 0, \frac{1}{2t}, -\frac{1}{2t} \right\}.$$

We then get the logarithmic terms Lt_2 and Lt_3 from τ_2 and τ_3 , respectively.

$$Lt_2 = (1-a)^2 \exp((\alpha-1)D_\alpha^0) + 2a(1-a) \exp((\alpha-1)D_\alpha^1) + a^2 \exp((\alpha-1)D_\alpha^2),$$

and

$$Lt_3 = (1-b)^2 \exp((\alpha-1)D_\alpha^1) + 2b(1-b) \exp((\alpha-1)D_\alpha^2) + b^2 \exp((\alpha-1)D_\alpha^3).$$

By substituting $a = \frac{1}{2} + b + \rho$ into Lt_2 , we decompose the difference between Lt_2 and Lt_3 as $h(b) = Lt_2 - Lt_3$, yielding

$$\begin{aligned} h(b) = & \underbrace{(1-3e^c+3e^{4c}-e^{9c})b^2}_A \\ & + \underbrace{((2\rho-1)+(2-4\rho)e^c+(2\rho-1)e^{4c})b}_B \\ & + \underbrace{(\rho-\frac{1}{2})^2-(2\rho^2+\frac{1}{2})e^c+(\rho+\frac{1}{2})^2e^{4c}}_C, \end{aligned}$$

where $c = \frac{\alpha(\alpha-1)}{2\sigma^2}$.

$$a = \frac{1}{2} + b + \rho$$

Substituting this into

$$\begin{aligned} Lt_2 &= (1-a)^2 + 2a(1-a)e^c + a^2e^{4c}, \\ Lt_3 &= (1-b)^2e^c + 2b(1-b)e^{4c} + b^2e^{9c}. \end{aligned}$$

we derive the explicit expression of

$$h(b) = Lt_2 - Lt_3.$$

We first expand each term involving a . Since

$$1 - a = 1 - \left(\frac{1}{2} + b + \rho\right) = \frac{1}{2} - b - \rho,$$

we obtain

$$\begin{aligned} (1-a)^2 &= \left(\frac{1}{2} - b - \rho\right)^2 \\ &= \frac{1}{4} - b - \rho + b^2 + 2b\rho + \rho^2 \\ &= b^2 - (1-2\rho)b + \left(\rho - \frac{1}{2}\right)^2. \end{aligned}$$

Next,

$$\begin{aligned} a(1-a) &= \left(\frac{1}{2} + b + \rho\right) \left(\frac{1}{2} - b - \rho\right) \\ &= \frac{1}{4} - b^2 - 2b\rho - \rho^2. \end{aligned}$$

Therefore,

$$2a(1-a)e^c = \left(\frac{1}{2} - 2b^2 - 4\rho b - 2\rho^2\right)e^c.$$

For the last term,

$$a^2 = \left(\frac{1}{2} + b + \rho\right)^2 = b^2 + (1+2\rho)b + \left(\rho + \frac{1}{2}\right)^2,$$

and hence

$$a^2 e^{4c} = \left[b^2 + (1+2\rho)b + \left(\rho + \frac{1}{2}\right)^2\right] e^{4c}.$$

Combining the above expansions yields

$$\begin{aligned} Lt_2 &= (1-a)^2 + 2a(1-a)e^c + a^2 e^{4c} \\ &= \left[b^2 - (1-2\rho)b + \left(\rho - \frac{1}{2}\right)^2\right] \\ &\quad + \left(\frac{1}{2} - 2b^2 - 4\rho b - 2\rho^2\right)e^c \\ &\quad + \left[b^2 + (1+2\rho)b + \left(\rho + \frac{1}{2}\right)^2\right] e^{4c}. \end{aligned}$$

We now expand Lt_3 :

$$(1-b)^2 e^c = (1-2b+b^2)e^c,$$

$$2b(1-b)e^{4c} = (2b-2b^2)e^{4c}.$$

Thus,

$$Lt_3 = (1-2b+b^2)e^c + (2b-2b^2)e^{4c} + b^2 e^{9c}.$$

Hence,

$$\begin{aligned} h(b) &= Lt_2 - Lt_3 \\ &= \left[b^2 - (1-2\rho)b + \left(\rho - \frac{1}{2}\right)^2\right] \\ &\quad + \left(\frac{1}{2} - 2b^2 - 4\rho b - 2\rho^2\right)e^c \\ &\quad + \left[b^2 + (1+2\rho)b + \left(\rho + \frac{1}{2}\right)^2\right] e^{4c} \\ &\quad - (1-2b+b^2)e^c - (2b-2b^2)e^{4c} - b^2 e^{9c}. \end{aligned}$$

We now collect terms according to the powers of b .

Quadratic term. The coefficient of b^2 is

$$b^2 - 2b^2 e^c + b^2 e^{4c} - b^2 e^c + 2b^2 e^{4c} - b^2 e^{9c} = (1-3e^c+3e^{4c}-e^{9c})b^2.$$

Linear term. The coefficient of b is

$$-(1-2\rho)b - 4\rho e^c b + (1+2\rho)e^{4c}b + 2e^c b - 2e^{4c}b,$$

which simplifies to

$$\left((2\rho-1) + (2-4\rho)e^c + (2\rho-1)e^{4c}\right)b.$$

Constant term. The constant term is

$$\left(\rho - \frac{1}{2}\right)^2 + \left(\frac{1}{2} - 2\rho^2\right)e^c + \left(\rho + \frac{1}{2}\right)^2 e^{4c} - e^c,$$

that is,

$$\left(\rho - \frac{1}{2}\right)^2 - \left(2\rho^2 + \frac{1}{2}\right)e^c + \left(\rho + \frac{1}{2}\right)^2 e^{4c}.$$

Therefore, we finally obtain

$$\begin{aligned} h(b) &= (1-3e^c+3e^{4c}-e^{9c})b^2 \\ &\quad + \left((2\rho-1) + (2-4\rho)e^c + (2\rho-1)e^{4c}\right)b \\ &\quad + \left(\rho - \frac{1}{2}\right)^2 - \left(2\rho^2 + \frac{1}{2}\right)e^c + \left(\rho + \frac{1}{2}\right)^2 e^{4c}. \end{aligned}$$

Let $g(e^c) = 1-3e^c+3e^{4c}-e^{9c}$ for $e^c > 1$, $g(e^c)$ is monotonically decreasing on $(1, \infty)$. Hence, we have $A < 0$. Similarly, we have $B < 0$.

Recall that

$$B = (2\rho-1) + (2-4\rho)e^c + (2\rho-1)e^{4c}.$$

we factorize B as

$$\begin{aligned} B &= (2\rho-1) - 2(2\rho-1)e^c + (2\rho-1)e^{4c} \\ &= (2\rho-1)(1-2e^c+e^{4c}) < 0, \end{aligned}$$

for $\rho < \frac{1}{2}$.

Since $h(0.5) < 0$ holds for all $t > 6$, it follows that $Lt_2 < Lt_3$ when h has no root in $[0, 0.5)$. To facilitate subsequent analysis, we set $b_0 = 0$ when h has no positive solution.

Let $x = e^c$, $h(0.5) = \rho^2(1-2x+x^4) + 2\rho(x^4-x) - \frac{1}{4}x^9 + \frac{1}{2}x^4 - \frac{1}{4}x$.
1) $\rho = 0$, $h(0.5) = -\frac{1}{4}(x^9 - 2x^4 + x)$. Let $g(x) = x^9 - 2x^4 + x$,
 $g'(x) = 9x^8 - 8x^3 + 1 > g'(1) = 0$, $g(x) > g(1) = 0$ for all $x > 1$.

Therefore, $h(0.5) < 0$.

2) $\rho = \frac{1}{2t}$. Define

$$\begin{aligned} D &= x^8 + x^7 + x^6 + x^5 + x^4 - x^3 - x^2 - x, \\ E &= x^3 + x^2 + x, \quad F = E - 1. \end{aligned}$$

$$\begin{aligned} h\left(\frac{1}{2}\right) &= \frac{1}{4t^2} \left[(1 - 2x + x^4) + 4t(x^4 - x) + t^2(-x^9 + 2x^4 - x) \right] \\ &= \frac{x-1}{4t^2} \left(-t^2D + 4tE + F \right). \end{aligned}$$

Let $\Phi(t) = -Dt^2 + 4tE + F$; we need $\Phi(t) < 0$. Since $D, E, F > 0$, this holds when

$$t > \frac{2E + \sqrt{4E^2 + DF}}{D} =: T(x).$$

Since $3D > 2E$ and $10D^2 > 4E^2 + DF$, we have $T(x) < 3 + \sqrt{10}$. Therefore, it suffices that $t > 3 + \sqrt{10} > 6$.

3) $\rho = -\frac{1}{2t}$.

$$\begin{aligned} h\left(\frac{1}{2}\right) &= \frac{1}{4t^2} \left[(1 - 2x + x^4) - 4t(x^4 - x) + t^2(-x^9 + 2x^4 - x) \right] \\ &= -\frac{x-1}{4t^2} \left(t^2D + 4tE - F \right) \\ &= -\frac{x-1}{4t^2} \left(t^2D + (4t-1)E + 1 \right) < 0, \end{aligned}$$

since the term in $[\cdot]$ is positive.

Otherwise, $h = 0$ has exactly one solution in the interval $[0, 0.5)$, which is derived as $b_0 = \frac{-B - \sqrt{B^2 - 4AC}}{2A}$. Consequently, if $\lceil \frac{n_s - t}{2} \rceil < b_0t$, namely, $n_s - t \leq 2\lceil b_0t \rceil - 2$, we have $Lt_2 > Lt_3$.

If $\lceil \frac{n_s - t}{2} \rceil < b_0t$, then, since the left-hand side is an integer, we have $\lceil \frac{n_s - t}{2} \rceil \leq \lceil b_0t \rceil - 1$. Hence $\frac{n_s - t}{2} \leq \lceil \frac{n_s - t}{2} \rceil \leq \lceil b_0t \rceil - 1$, which implies $n_s - t \leq 2\lceil b_0t \rceil - 2$. $\square$

3 Proof of Sensitivity characterization of spaced sampling.

THEOREM 4.3 (SENSITIVITY CHARACTERIZATION OF SPACED SAMPLING.). *Given a batch size K and a set of segments $\mathcal{S}$ obtained by partitioning the time series chronologically, let $t = \lfloor |\mathcal{S}|/K \rfloor \geq 7$, and let n_s denote the number of w -span segments. Define $c = \frac{\alpha(\alpha-1)}{2\sigma^2}$, where α is the order of Rényi divergence and σ is the noise multiplier of the Gaussian mechanism under spaced sampling. Let $A = 1 - 3e^c + 3e^{4c} - e^{9c}$, $B = (2\rho - 1) + (2 - 4\rho)e^c + (2\rho - 1)e^{4c}$, $C = \left(\rho - \frac{1}{2}\right)^2 - \left(2\rho^2 + \frac{1}{2}\right)e^c + \left(\rho + \frac{1}{2}\right)^2e^{4c}$, where $\rho \in \{0, \frac{1}{2t}, -\frac{1}{2t}\}$ is the discretization error induced by the ceiling operation. Define*

$$b_0 = \begin{cases} 0.5, & \text{if } n_s < t, \\ b^*, & \text{if } n_s \geq t, B^2 - 4AC \geq 0, \text{ and } b^* = \frac{-B - \sqrt{B^2 - 4AC}}{2A} \in [0, \frac{1}{2}), \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Then the privacy loss attains its maximum value when

$\max\left(\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor, 0\right) + 2$ w -span segments are included in one batch.

PROOF. In Case1, at most two neighboring windows can contain w -span segments. Hence, the maximum number of included w -span segments is 2. On the other hand, since $n_s < t$, we have $n_s - (2\lceil b_0t \rceil - 1) < t$, which implies $\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor \leq 0$. Therefore, $\max\left(\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor, 0\right) + 2 = 2$. The remaining two cases can be verified similarly.

Case 2: $t \leq n_s < 2t$. As shown above, let $b = \frac{1}{t} \lceil \frac{n_s - t}{2} \rceil$. If $b < b_0$, then $Lt_2 > Lt_3$, and the privacy loss is maximized when the w -span segments are evenly split between two windows. Otherwise, the privacy loss is maximized when one window is first filled with w -span segments and the remaining w -span segments are evenly split between the two adjacent windows. Moreover, $\lceil \frac{n_s - t}{2} \rceil < b_0t$ if and only if $n_s - t \leq 2\lceil b_0t \rceil - 2$. Hence, if $n_s - t \leq 2\lceil b_0t \rceil - 2$, the maximum number of included w -span segments is 2; otherwise, it is 3.

These two cases can be unified as $\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor + 2$. Indeed, since $t \leq n_s < 2t$, we have $0 \leq n_s - t < t$. If $n_s - t \leq 2\lceil b_0t \rceil - 2$, then $n_s - (2\lceil b_0t \rceil - 1) < t$, and thus $\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor = 0$, which gives 2. Otherwise, $n_s - t \geq 2\lceil b_0t \rceil - 1$, and hence $n_s - (2\lceil b_0t \rceil - 1) \geq t$. Since $n_s < 2t$, we also have $n_s - (2\lceil b_0t \rceil - 1) < 2t$, so $\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor = 1$, which gives 3.

Case 3: $n_s \geq 2t$. Let $n = \lfloor \frac{n_s}{t} \rfloor$ and write $n_s = (n-1)t + (t+r)$, where $0 \leq r < t$. After filling $(n-1)$ windows completely, the remaining $t+r$ w -span segments fall into the regime of Case 2. Therefore, if $r \leq 2\lceil b_0t \rceil - 2$, the remaining segments are evenly split between two windows, and the maximum number of included w -span segments is $(n-1) + 2 = n+1$. Otherwise, one window is first filled with w -span segments and the remaining segments are evenly split between the two adjacent windows, in which case the maximum number of included w -span segments is $n+2$.

These two subcases can again be unified. Since $n_s = nt + r$, we have $\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor = \left\lfloor \frac{nt + r - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor$. If $r \leq 2\lceil b_0t \rceil - 2$, then $r - (2\lceil b_0t \rceil - 1) < 0$, and thus $\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor = n-1$, which yields $\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor + 2 = n+1$. Otherwise, $r \geq 2\lceil b_0t \rceil - 1$, and hence $0 \leq r - (2\lceil b_0t \rceil - 1) < t$. Therefore, $\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor = n$, which gives $\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor + 2 = n+2$.

Combining the above three cases, the privacy loss attains its maximum value when $\max\left(\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor, 0\right) + 2$ w -span segments are included in one batch. $\square$

4 Comparison with Poisson Sampling

THEOREM 4.4. *Given batch size K and segment count $|\mathcal{S}|$, let $\mathbb{E}_{\text{Poisson}} = n_s q = n_s \frac{K}{|\mathcal{S}|}$ denote the expected number of sampled w -span segments under Poisson sampling. Then the number of included w -span segments under spaced sampling is upper bounded by $\frac{\mathbb{E}_{\text{Poisson}} |\mathcal{S}|}{|\mathcal{S}| - K} + 3$.*

PROOF. The number of included w -span segments under spaced sampling is upper bounded by $\max\left(\left\lfloor \frac{n_s - (2\lceil b_0t \rceil - 1)}{t} \right\rfloor, 0\right) + 2 \leq$

$$\left\lfloor \frac{n_s+1}{t} \right\rfloor + 2 \leq \frac{n_s+1}{t} + 2.$$

Since $t = \lfloor |S|/K \rfloor \geq |S|/K - 1$, we have $\frac{1}{t} \leq \frac{K}{|S|-K}$. Therefore, $\frac{n_s+1}{t} + 2 \leq \frac{n_s K}{|S|-K} + 3 = \mathbb{E}_{\text{Poisson}} \frac{|S|}{|S|-K} + 3$, where $\mathbb{E}_{\text{Poisson}}$ denotes the expectation of included w -span segments under Poisson sampling, i.e., $\mathbb{E}_{\text{Poisson}} = n_s \cdot \frac{K}{|S|}$. $\square$

5 Comparison of Covariances

THEOREM 5.2. *Let $\hat{g}_{\text{psp}}$ denote the resulting gradient estimator under Poisson spaced sampling, and let $\hat{g}_{\text{poi}}$ denote the gradient estimator under Poisson sampling with the same expected batch size. Then, we have*

$$\text{Cov}(\hat{g}_{\text{poi}}) - \text{Cov}(\hat{g}_{\text{psp}}) \succeq 0.$$

PROOF. Let $\mathcal{S}$ be the set of time series segments, K be the batch size, and λ be the window scaling factor. Poisson spaced sampling partitions the time axis into $N = \lfloor K/\lambda \rfloor$ consecutive windows and lets $\mathcal{S}_k$ denote the set of segments falling into the k -th window with $k \in \{1, \dots, N\}$. For each window $k \in \{1, \dots, N\}$, let g_k denote the gradient of a segment sampled from $\mathcal{S}_k$ according to the within-window sampling rule. Define

$$\mu_k = \mathbb{E}[g_k], \quad \Sigma_k = \text{Cov}(g_k), \quad \mu = \frac{1}{N} \sum_{k=1}^N \mu_k.$$

Define the within-window and between-window covariances by

$$\Sigma_{WW} = \frac{1}{N} \sum_{k=1}^N \Sigma_k, \quad \Sigma_{BW} = \frac{1}{N} \sum_{k=1}^N (\mu_k - \mu)(\mu_k - \mu)^\top.$$

By the law of total covariance,

$$\text{Cov}(g) = \frac{1}{N} \sum_{k=1}^N \Sigma_k + \frac{1}{N} \sum_{k=1}^N (\mu_k - \mu)(\mu_k - \mu)^\top = \Sigma_{WW} + \Sigma_{BW}.$$

By construction, a segment gradient g drawn uniformly from the union of the N windows follows the mixture distribution that first chooses a window uniformly from $\{1, \dots, N\}$ and then draws a segment according to the within-window sampling rule. Hence,

$$\mu = \mathbb{E}[g] = \frac{1}{N} \sum_{k=1}^N \mu_k.$$

We now decompose $\text{Cov}(g)$ into within-window and between-window parts. Using the definition of covariance,

$$\text{Cov}(g) = \mathbb{E}[(g - \mu)(g - \mu)^\top].$$

For a sample drawn from window k , we write

$$g - \mu = (g_k - \mu_k) + (\mu_k - \mu).$$

Substituting this into the covariance expression and averaging over the N windows gives

$$\text{Cov}(g) = \frac{1}{N} \sum_{k=1}^N \mathbb{E}[(g_k - \mu_k) + (\mu_k - \mu)((g_k - \mu_k) + (\mu_k - \mu))^\top].$$

Expanding the product,

$$\begin{aligned} \text{Cov}(g) &= \frac{1}{N} \sum_{k=1}^N \mathbb{E}[(g_k - \mu_k)(g_k - \mu_k)^\top] \\ &\quad + \frac{1}{N} \sum_{k=1}^N \mathbb{E}[(g_k - \mu_k)(\mu_k - \mu)^\top] \\ &\quad + \frac{1}{N} \sum_{k=1}^N \mathbb{E}[(\mu_k - \mu)(g_k - \mu_k)^\top] \\ &\quad + \frac{1}{N} \sum_{k=1}^N \mathbb{E}[(\mu_k - \mu)(\mu_k - \mu)^\top]. \end{aligned}$$

Since $\mu_k - \mu$ is deterministic and $\mathbb{E}[g_k - \mu_k] = 0$, the two cross terms vanish:

$$\mathbb{E}[(g_k - \mu_k)(\mu_k - \mu)^\top] = \mathbb{E}[g_k - \mu_k](\mu_k - \mu)^\top = 0,$$

and similarly,

$$\mathbb{E}[(\mu_k - \mu)(g_k - \mu_k)^\top] = 0.$$

Therefore,

$$\begin{aligned} \text{Cov}(g) &= \frac{1}{N} \sum_{k=1}^N \text{Cov}(g_k) + \frac{1}{N} \sum_{k=1}^N (\mu_k - \mu)(\mu_k - \mu)^\top \\ &= \frac{1}{N} \sum_{k=1}^N \Sigma_k + \frac{1}{N} \sum_{k=1}^N (\mu_k - \mu)(\mu_k - \mu)^\top \\ &= \Sigma_{WW} + \Sigma_{BW}. \end{aligned}$$

(1) Analyze the variance of Poisson spaced sampling. To simplify the analysis, we ignore the effect of rounding and set λ as the probability of selecting a window. Under Poisson spaced sampling, one segment is drawn independently from each selected window. Let $I_k \sim \text{Bernoulli}(\lambda)$ be the inclusion indicator for window selection, let N denote the random batch size with $\mathbb{E}[\lambda N] = K$, and $\hat{g}_{\text{psp}} = \frac{1}{K} \sum I_k g_k$.

$$\begin{aligned} \text{Cov}(\hat{g}_{\text{psp}}) &= \text{Cov}\left(\frac{1}{K} \sum_{k=1}^N I_k g_k\right) = \frac{1}{K^2} \sum_{k=1}^N \text{Cov}(I_k g_k) \\ &= \frac{1}{K^2} \sum_{k=1}^N \mathbb{E}[(I_k g_k)(I_k g_k)^\top] - \mathbb{E}[I_k g_k] \mathbb{E}[I_k g_k]^\top \\ &= \frac{1}{K} \Sigma_{WW} + \frac{1-\lambda}{K} \Sigma_{BW} + \frac{1-\lambda}{K} \mu \mu^\top. \end{aligned}$$

$$\begin{aligned}
 \text{Cov}(\hat{g}_{\text{psp}}) &= \text{Cov}\left(\frac{1}{K} \sum_{k=1}^N I_k g_k\right) \\
 &= \frac{1}{K^2} \sum_{k=1}^N \text{Cov}(I_k g_k) \\
 &= \frac{1}{K^2} \sum_{k=1}^N (\mathbb{E}[(I_k g_k)(I_k g_k)^\top] - \mathbb{E}[I_k g_k] \mathbb{E}[I_k g_k]^\top) \\
 &= \frac{1}{K^2} \sum_{k=1}^N (\lambda \mathbb{E}[g_k g_k^\top] - \lambda^2 \mu_k \mu_k^\top) \\
 &= \frac{1}{K^2} \sum_{k=1}^N (\lambda \text{Cov}(g_k) + \lambda \mathbb{E}(g_k) \mathbb{E}(g_k)^\top - \lambda^2 \mu_k \mu_k^\top) \\
 &= \frac{1}{K^2} \sum_{k=1}^N (\lambda \text{Cov}(g_k) + \lambda(1 - \lambda) \mu_k \mu_k^\top)
 \end{aligned}$$

Since $\mu = \mathbb{E}[g] = \frac{1}{N} \sum_{k=1}^N \mu_k = \frac{1}{K} \sum_{k=1}^N \lambda \mu_k$ and $\Sigma_{BW} = \frac{1}{N} \sum_{k=1}^N (\mu_k - \mu)(\mu_k - \mu)^\top = \frac{1}{N} \sum_{k=1}^N \mu_k \mu_k^\top - \mu \mu^\top = \frac{1}{N} \sum_{k=1}^N \mu_k \mu_k^\top - \mu \mu^\top$, the equation above can be rewritten as

$$\begin{aligned}
 &= \frac{1}{K} \Sigma_{WW} + \frac{1 - \lambda}{K} (\Sigma_{BW} + \mu \mu^\top) \\
 &= \frac{1}{K} \Sigma_{WW} + \frac{1 - \lambda}{K} \Sigma_{BW} + \frac{1 - \lambda}{K} \mu \mu^\top.
 \end{aligned}$$

(2) Analyze the variance of Poisson sampling. Let $I_s \sim \text{Bernoulli}(q)$ be the inclusion indicator, let M denote the random batch size with $\mathbb{E}[M] = K$, and $\hat{g}_{\text{poi}} = \frac{1}{K} \sum_{s \in \mathcal{S}} I_s g_s$. where, conditional on M , the sampled gradients $\tilde{g}_1, \dots, \tilde{g}_M$ follow the distribution of g . Then,

$$\begin{aligned}
 \text{Cov}(\hat{g}_{\text{poi}}) &= \frac{1}{K^2} \sum_{s \in \mathcal{S}} \text{Cov}(I_s g_s) = \frac{|S|}{K^2} (q \text{Cov}(g) + q(1 - q) \mu \mu^\top) \\
 &= \frac{|S|}{K^2} (q(\Sigma_{WW} + \Sigma_{BW}) + q(1 - q) \mu \mu^\top) \\
 &= \frac{1}{K} (\Sigma_{WW} + \Sigma_{BW}) + \frac{1 - q}{K} \mu \mu^\top,
 \end{aligned}$$

where we use $\text{Cov}(g) = \Sigma_{WW} + \Sigma_{BW}$.

$$\begin{aligned}
 \text{Cov}(\hat{g}_{\text{poi}}) &= \text{Cov}\left(\frac{1}{K} \sum_{s \in \mathcal{S}} I_s g_s\right) = \frac{1}{K^2} \sum_{s \in \mathcal{S}} \text{Cov}(I_s g_s) \\
 &= \frac{1}{K^2} \sum_{s \in \mathcal{S}} \mathbb{E}[(I_s g_s)(I_s g_s)^\top] - \mathbb{E}[I_s g_s] \mathbb{E}[I_s g_s]^\top
 \end{aligned}$$

Since $I_s \in \{0, 1\}$, we have $I_s^2 = I_s$. Hence, $\mathbb{E}[(I_s g_s)(I_s g_s)^\top] = \mathbb{E}[I_s g_s g_s^\top] = \mathbb{E}[I_s] \mathbb{E}[g_s g_s^\top] = q \mathbb{E}[g g^\top] = q(\text{Cov}(g) + \mu \mu^\top)$. And

$\mathbb{E}[I_s g_s] = \mathbb{E}[I_s] \mathbb{E}[g_s] = q \mu$. Therefore,

$$\begin{aligned}
 &\frac{1}{K^2} \sum_{s \in \mathcal{S}} \mathbb{E}[(I_s g_s)(I_s g_s)^\top] - \mathbb{E}[I_s g_s] \mathbb{E}[I_s g_s]^\top \\
 &= \frac{|S|}{K^2} [q(\text{Cov}(g) + \mu \mu^\top) - q^2 \mu \mu^\top] \\
 &= \frac{|S|}{K^2} [q \text{Cov}(g) + q(1 - q) \mu \mu^\top] \\
 &= \frac{1}{K} (\Sigma_{WW} + \Sigma_{BW}) + \frac{1 - q}{K} \mu \mu^\top.
 \end{aligned}$$

Finally, we **compare the variance between Poisson spaced sampling and Poisson sampling**,

$$\text{Cov}(\hat{g}_{\text{poi}}) - \text{Cov}(\hat{g}_{\text{psp}}) = \frac{\lambda}{K} \Sigma_{BW} + \frac{\lambda - q}{K} \mu \mu^\top \succeq 0,$$

since Σ_{BW} is positive semidefinite and $\mu \mu^\top \succeq 0$. $\square$

6 Convergence Analysis

Convergence Analysis. We assume that the loss function f is L -smooth in time series training, and f is bounded below as $f^* = \inf_{\theta} f(\theta)$. Given the gradients g_i , the noisy clipped gradient of the j -th update is defined as $\tilde{g}_j = \frac{1}{K} \left(\sum_{i=1}^M \tilde{g}_i + z_j \right)$, where $z_j \sim \mathcal{N}(0, 4C^2 \sigma^2 \mathbf{I})$ and $\tilde{g}_i$ represent the clipped gradients. Let $\tilde{g}_j = \nabla f(\theta_j) + b_j + v_j$, where b_j denotes the bias induced by gradient clipping, and v_j is a zero-mean noise term capturing both the added Gaussian noise and the intrinsic randomness of the gradient estimator. We use a standard first-order measure, the average expected squared gradient norm, as a stationarity measure for nonconvex optimization. According to the Descent Lemma, the average expected squared gradient norm after J steps is derived with learning rate $\eta < \frac{1}{4L}$ (L represents the Lipschitz constant) as

$$\begin{aligned}
 \frac{1}{J} \sum_{j=0}^{J-1} \mathbb{E} \|\nabla f(\theta_j)\|^2 &\leq \frac{4(f(\theta_0) - f^*)}{\eta J} + \frac{3}{J} \sum_{j=0}^{J-1} \mathbb{E} \|b_j\|^2 \\
 &\quad + 2L\eta(\sigma_s + \sigma_N),
 \end{aligned}$$

where θ_j represents the parameters in the j -th update, $\sigma_N = \frac{4C^2 \sigma^2 \text{tr}(\mathbf{I})}{K^2}$, and $\sigma_s = \sup_t \text{tr}(\text{Cov}(\frac{1}{K} \sum_{i=1}^M g_i))$. Therefore, the convergence is related to the covariance of the sampling method. Since the covariance matrix is positive semidefinite, it follows that the σ_s under Poisson spaced sampling is smaller than that under Poisson sampling. Consequently, the convergence analysis suggests that Poisson spaced sampling may achieve faster convergence, whereas Poisson sampling may converge more slowly in this bound.

PROOF. Assume that f is L -smooth, namely

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2, \quad \forall x, y.$$

Let the parameter update be

$$\theta_{j+1} = \theta_j - \eta \hat{g}_j,$$

where the noisy clipped gradient estimator is defined as

$$\hat{g}_j = \frac{1}{K} \left(\sum_{i=1}^m \tilde{g}_i + z_j \right), \quad z_j \sim \mathcal{N}(0, 4C^2 \sigma^2 \mathbf{I}).$$

We decompose $\hat{g}_j$ as

$$\hat{g}_j = \nabla f(\theta_j) + b_j + v_j,$$

where b_j is the bias induced by gradient clipping, and v_j is the zero-mean fluctuation term capturing both the stochasticity of the gradient estimator and the injected Gaussian noise. Let $\mu_j = \mathbb{E}[\hat{g}_j \mid \theta_j]$. Then define $b_j = \mu_j - \nabla f(\theta_j)$, $v_j = \hat{g}_j - \mu_j$. We have

$$\begin{aligned}\mathbb{E}[v_j \mid \theta_j] &= \mathbb{E}[\hat{g}_j - \mu_j \mid \theta_j] \\ &= \mathbb{E}[\hat{g}_j - \mathbb{E}[\hat{g}_j \mid \theta_j] \mid \theta_j] = 0.\end{aligned}$$

By L -smoothness,

$$\begin{aligned}f(\theta_{j+1}) &\leq f(\theta_j) + \langle \nabla f(\theta_j), \theta_{j+1} - \theta_j \rangle \\ &\quad + \frac{L}{2} \|\theta_{j+1} - \theta_j\|^2 \\ &= f(\theta_j) - \eta \langle \nabla f(\theta_j), \hat{g}_j \rangle \\ &\quad + \frac{L\eta^2}{2} \|\hat{g}_j\|^2.\end{aligned}$$

Taking conditional expectation given θ_j ,

$$\begin{aligned}\mathbb{E}[f(\theta_{j+1}) \mid \theta_j] &\leq f(\theta_j) \\ &\quad - \eta \mathbb{E}[\langle \nabla f(\theta_j), \hat{g}_j \rangle \mid \theta_j] \\ &\quad + \frac{L\eta^2}{2} \mathbb{E}[\|\hat{g}_j\|^2 \mid \theta_j].\end{aligned}$$

As for the inner product term, we have

$$\begin{aligned}\mathbb{E}[\langle \nabla f(\theta_j), \hat{g}_j \rangle \mid \theta_j] &= \|\nabla f(\theta_j)\|^2 \\ &\quad + \langle \nabla f(\theta_j), b_j \rangle \\ &\quad + \mathbb{E}[\langle \nabla f(\theta_j), v_j \rangle \mid \theta_j] \\ &= \|\nabla f(\theta_j)\|^2 + \langle \nabla f(\theta_j), b_j \rangle,\end{aligned}$$

since $\mathbb{E}[v_j \mid \theta_j] = 0$. For the second moment term, we have

$$\begin{aligned}\mathbb{E}[\|\hat{g}_j\|^2 \mid \theta_j] &= \|\nabla f(\theta_j) + b_j\|^2 \\ &\quad + 2 \langle \nabla f(\theta_j) + b_j, \mathbb{E}[v_j \mid \theta_j] \rangle \\ &\quad + \mathbb{E}[\|v_j\|^2 \mid \theta_j] \\ &= \|\nabla f(\theta_j) + b_j\|^2 + \mathbb{E}[\|v_j\|^2 \mid \theta_j] \\ &\leq 2 \|\nabla f(\theta_j)\|^2 + 2 \|b_j\|^2 \\ &\quad + \mathbb{E}[\|v_j\|^2 \mid \theta_j].\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbb{E}[f(\theta_{j+1}) \mid \theta_j] &\leq f(\theta_j) - \eta \|\nabla f(\theta_j)\|^2 - \eta \langle \nabla f(\theta_j), b_j \rangle + L\eta^2 \|\nabla f(\theta_j)\|^2 \\ &\quad + L\eta^2 \|b_j\|^2 + \frac{L\eta^2}{2} \mathbb{E}[\|v_j\|^2 \mid \theta_j] \\ &\leq f(\theta_j) - \eta \|\nabla f(\theta_j)\|^2 + \frac{\eta}{2} \|\nabla f(\theta_j)\|^2 + \frac{\eta}{2} \|b_j\|^2 \\ &\quad + L\eta^2 \|\nabla f(\theta_j)\|^2 + L\eta^2 \|b_j\|^2 + \frac{L\eta^2}{2} \mathbb{E}[\|v_j\|^2 \mid \theta_j] \\ &= f(\theta_j) - \frac{\eta}{2} \|\nabla f(\theta_j)\|^2 \\ &\quad + \frac{\eta}{2} \|b_j\|^2 + L\eta^2 \|\nabla f(\theta_j)\|^2 \\ &\quad + L\eta^2 \|b_j\|^2 + \frac{L\eta^2}{2} \mathbb{E}[\|v_j\|^2 \mid \theta_j]\end{aligned}$$

Then, we have

$$\begin{aligned}\mathbb{E}[f(\theta_{j+1}) \mid \theta_j] &\leq f(\theta_j) - \eta \left(\frac{1}{2} - L\eta \right) \|\nabla f(\theta_j)\|^2 \\ &\quad + \left(\frac{\eta}{2} + L\eta^2 \right) \|b_j\|^2 \\ &\quad + \frac{L\eta^2}{2} \mathbb{E}[\|v_j\|^2 \mid \theta_j].\end{aligned}$$

Taking total expectation, we have

$$\begin{aligned}\mathbb{E}[f(\theta_{j+1})] &\leq \mathbb{E}[f(\theta_j)] - \eta \left(\frac{1}{2} - L\eta \right) \mathbb{E}[\|\nabla f(\theta_j)\|^2] \\ &\quad + \left(\frac{\eta}{2} + L\eta^2 \right) \mathbb{E}[\|b_j\|^2] + \frac{L\eta^2}{2} \mathbb{E}[\|v_j\|^2].\end{aligned}$$

Let $\eta < \frac{1}{4L}$. Then, we have

$$\begin{aligned}\mathbb{E}[f(\theta_{j+1})] &\leq \mathbb{E}[f(\theta_j)] - \frac{\eta}{4} \mathbb{E}[\|\nabla f(\theta_j)\|^2] \\ &\quad + \frac{3\eta}{4} \mathbb{E}[\|b_j\|^2] + \frac{L\eta^2}{2} \mathbb{E}[\|v_j\|^2].\end{aligned}$$

Summing over $t = 0, 1, \dots, J-1$, we have

$$\begin{aligned}\mathbb{E}[f(\theta_J)] - \mathbb{E}[f(\theta_0)] &\leq -\frac{\eta}{4} \sum_{t=0}^{J-1} \mathbb{E}[\|\nabla f(\theta_t)\|^2] \\ &\quad + \frac{3\eta}{4} \sum_{t=0}^{J-1} \mathbb{E}[\|b_t\|^2] \\ &\quad + \frac{L\eta^2}{2} \sum_{t=0}^{J-1} \mathbb{E}[\|v_t\|^2].\end{aligned}$$

Consequently,

$$\begin{aligned}\frac{1}{J} \sum_{t=0}^{J-1} \mathbb{E}[\|\nabla f(\theta_t)\|^2] &\leq \frac{4(f(\theta_0) - f^*)}{\eta J} \\ &\quad + \frac{3}{J} \sum_{t=0}^{J-1} \mathbb{E}[\|b_t\|^2] + 2L\eta\Sigma^2,\end{aligned}$$

where $\Sigma^2 = \sup_j \mathbb{E}[\|v_j\|^2]$, and $f^* = \inf_{\theta} f(\theta)$. Since $\mathbb{E}[v_j] = 0$, we have

$$\begin{aligned}\mathbb{E}[\|v_j\|^2] &= \mathbb{E} \text{tr}(v_j v_j^\top) \\ &= \text{tr} \mathbb{E}(v_j v_j^\top) = \text{Cov}(v_j).\end{aligned}$$

Since v_j captures the stochasticity of the gradient estimator together with the injected Gaussian noise, we further bound

$$\Sigma^2 \leq \text{Cov}(v_j) = \sigma_s + \sigma_N,$$

where $\sigma_s = \sup_j \text{tr} \left(\text{Cov} \left(\frac{1}{K} \sum_{i=1}^M g_i \right) \right)$ denotes the variance contribution of the stochastic gradient estimator, and $\sigma_N = \frac{4C^2\sigma^2}{K^2} \text{tr}(I)$ is the variance of the injected Gaussian noise. Therefore,

$$\begin{aligned}\frac{1}{J} \sum_{t=0}^{J-1} \mathbb{E}[\|\nabla f(\theta_t)\|^2] &\leq \frac{4(f(\theta_0) - f^*)}{\eta J} \\ &\quad + \frac{3}{J} \sum_{t=0}^{J-1} \mathbb{E}[\|b_t\|^2] \\ &\quad + 2L\eta(\sigma_s + \sigma_N).\end{aligned}$$

□

C. ADDITIONAL EXPERIMENTAL RESULTS

1 Concept Drift

To examine the effect of concept drift, we conduct additional experiments on the LargeST Benchmark Dataset for forecasting and the USC-SIPI Human Activity Dataset for classification, which exhibit relatively weak and strong temporal distribution shifts, respectively. Following [28], we characterize the effect of concept drift using the covariance of stochastic gradients. The results show that the gradient covariance varies substantially less across temporal regimes on the LargeST Benchmark Dataset than on the USC-SIPI Human Activity Dataset, consistent with the weaker distribution shift in LargeST. Moreover, across both datasets, Poisson spaced sampling yields substantially smaller gradient covariance than Poisson sampling, suggesting that its control over the number of sampled w -span segments per batch reduces the impact of concept drift on optimization.

Table 2: Comparison of gradient covariance on the LargeST Benchmark Dataset.

Metric	Poisson Sampling	Poisson Spaced Sampling	Reduction
Trace	0.039934	0.037881	5.14%
Frobenius	0.026333	0.024182	8.17%
Spectral	0.024909	0.022690	8.91%

Table 3: Comparison of gradient covariance on the USC-SIPI Human Activity Dataset.

Metric	Poisson Sampling	Poisson Spaced Sampling	Reduction
Trace	14.346055	9.543125	33.48%
Frobenius	5.217137	3.288386	36.97%
Spectral	3.705133	2.159506	41.72%

The covariance of gradients is a matrix, and thus we quantify its magnitude using three scalar metrics: the trace, Frobenius norm, and spectral norm. Let $\hat{g}$ denote the stochastic gradient and $\mu = \mathbb{E}[\hat{g}]$ denote its expectation. The gradient covariance matrix is defined as $C = \mathbb{E}[(\hat{g} - \mu)(\hat{g} - \mu)^\top]$. We compute the trace $\text{tr}(C) = \sum_i \lambda_i = \mathbb{E}\|\hat{g} - \mu\|^2$, the Frobenius norm $\|C\|_F = \sqrt{\sum_{i,j} C_{ij}^2} = \sqrt{\sum_i \lambda_i^2}$, and the spectral norm $\|C\|_2 = \lambda_{\max}(C) = \max_{\|v\|_2=1} \text{Var}(v^\top \hat{g})$, which characterize the overall gradient variance, the global covariance magnitude, and the maximum directional fluctuation, respectively. Smaller values of these metrics indicate reduced gradient dispersion and more stable gradient behavior, while larger values suggest stronger gradient instability and a stronger impact of concept drift on the optimization process.

To isolate the impact of sampling methods on concept drift, we train the PatchTST models for forecasting and classification without differential privacy. Both tasks use an segment length $\ell = 160$ and batch size 50. Specifically, the LargeST Benchmark Dataset contains 371 channels with prediction length $H = 20$ and MSE loss, while USC contains 6 channels, 12 classes, and cross-entropy loss. For

selecting the window scaling factor λ , we use $\epsilon = 6$, set w to 0.5% of the time series length, and set $\sigma = 5$. These parameters are used solely for λ selection, while the models are trained without DP noise. After selecting the best validation checkpoint, we fix the model parameters and compute per-sample gradients. We compare Poisson sampling and Poisson spaced sampling with an expected batch size $K = 60$ (unrelated to the batch size used for model training), where sampled gradients are averaged within each batch. For the trace metric, we analytically compute $\text{tr}(\text{Cov}(\hat{g})) = \mathbb{E}\|\hat{g}\|_2^2 - \|\mathbb{E}\hat{g}\|_2^2$ directly from individual gradients without constructing the full covariance matrix. Since constructing the covariance matrix is prohibitively expensive in high-dimensional gradient spaces, we apply CountSketch to project gradients into a 128-dimensional space. We then estimate the projected covariance matrix using 10K repeated samplings and compute the Frobenius and spectral norms in the compressed space. The repeated sampling only reduces Monte Carlo estimation error without additional retraining. Finally, we report the trace in the original gradient space and aggregate the Frobenius and spectral norms over 30 projection seeds.

2 Experimental Settings for Additional Scenarios

We first discuss the tasks clustering, change-point detection, anomaly detection and clarify how our sampling mechanism can be applied to these settings. To further demonstrate the generalizability of our approach beyond forecasting and classification, we also provide additional experimental results on anomaly detection.

- **Clustering.** For clustering, Poisson spaced sampling can be directly applied when fixed-length temporal segments are treated as clustering units and the clustering objective can be decomposed into per-segment contributions. For methods that rely on pairwise relations, global affinity, or density structure, it is not directly compatible with the current framework, but may be indirectly adapted through private representation learning.
- **Change-point detection.** For change-point detection, Poisson spaced sampling can be directly applied when the task is formulated as segment-level prediction, where each segment produces a local change score or change-point label. For methods that rely on continuous neighboring segments, cumulative statistics, or global temporal structure, it is not directly compatible with the current framework, but may be indirectly adapted by first learning private local representations or change scores and then performing change-point detection on them.
- **Anomaly detection.** For anomaly detection, Poisson spaced sampling can be directly applied when anomalies are detected from fixed-length segments using per-segment objectives, such as reconstruction or prediction errors. For timestamp-level detection, segment-level anomaly scores can be aggregated back to the time axis, while methods that rely on global or cumulative temporal statistics require indirect adaptation.

We evaluate multivariate time-series anomaly detection on machine 2-3 from the Server Machine Dataset (SMD) [56], where each time point contains 38 variables. The training and test sequences

each contain about 23K time points; the test sequence is chronologically split into 11,844 development points, a 400-point isolation gap, and 11,445 confirmation points, including 135 anomalies (1.18%). We use a feature autoencoder, $\text{Linear}(38, 4) \rightarrow \text{GELU} \rightarrow \text{Linear}(4, 38)$, and construct overlapping training segments with length $\ell = 250$ and stride $\gamma = 130$. Poisson sampling, structured sampling, and Poisson spaced sampling use the same configuration with batch size $K = 50$, noise multiplier $\sigma = 10$, and clipping norm $C = 0.05$. Anomalous points constitute only a small fraction of the test sequence, leading to a highly imbalanced evaluation setting. We therefore report point-wise Average Precision (AP) [54, 55] as the evaluation metric. The results are shown in Fig. 5. Poisson spaced sampling consistently outperforms the other two methods. At larger privacy budgets, Poisson sampling achieves higher AP than structured sampling. With only 181 candidate segments, structured sampling introduces more repeated segments within each update, whereas Poisson sampling retains more distinct segments and thus makes more effective use of the limited training data.

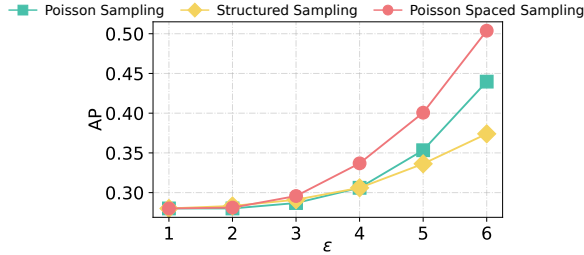


Figure 5: Anomaly detection on SMD dataset with segment length $\ell = 250$, stride $\gamma = 130$, batch size $K = 50$, noise multiplier $\sigma = 10$, and clipping norm $C = 0.05$.

3 Selection of the Window Scaling Factor λ

Efficiency of the selection process. In this part, we explain the selection of the window scaling factor λ and the RDP order α . Thank you for pointing out the potential computational overhead of this parameter selection. In practice, the overhead is small. Most quantities associated with different RDP orders can be precomputed and evaluated in parallel using vectorized operations. Moreover, the search is performed only once over a small set of candidate values. The selection of the RDP order α also follows standard practice in RDP accounting. Different Rényi orders α yield different converted privacy bounds, while the optimal order generally does not admit a closed-form solution. Therefore, RDP accountants typically evaluate a predefined set of orders and select the one that gives the smallest privacy loss [2, 31]. We follow the same principle for selecting λ and α , evaluating a small set of candidate values and choosing the pair that minimizes the resulting privacy loss. To quantify the actual overhead, we measure the runtime of privacy accounting and parameter selection using the experimental configurations ($\epsilon=6$), with each configuration repeated 100 times. The measurements are conducted on a server running Ubuntu 22.04.5 LTS with 144 CPU cores (Intel(R) Xeon(R) Platinum 8352V @ 2.10GHz). The results are shown below.

Table 4: Runtime of privacy accounting and parameter selection under different time series lengths.

Time-series length	Segment length	Stride	Mean runtime (s)
500K	100	50	0.812
525K	100	50	0.814
1.92M	200	50	0.880
2.60M	200	50	0.879

As shown in Table 4, the selection procedure takes less than one second, even for time series with millions of timestamps. The runtime generally increases with the time series length, although a slight non-monotonicity appears between the 1.92M and 2.60M cases. This is caused by rounding operations in the λ selection process: the 2.60M case results in a slightly larger window size and therefore requires fewer summation terms in the privacy accounting. Overall, the grid search incurs negligible overhead compared with neural network training. Fig. 6 further confirms this trend over a wider range of time series lengths.

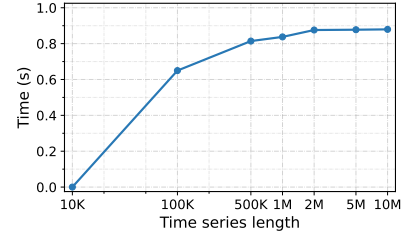


Figure 6: Runtime of parameter selection (λ and α) under different time series lengths with segment length $\ell = 200$, stride $\gamma = 50$, noise multiplier $\sigma = 5$, and $\epsilon = 6$.

4 Varying stride γ

As discussed in the experimental section, Poisson sampling requires a sufficiently large privacy budget to initiate training. Therefore, we evaluate the effect of γ on the LargeST Benchmark Dataset forecasting task with $\epsilon = 5$. As shown in Fig. 7, Poisson spaced sampling consistently achieves the lowest and most stable MSE across different stride values, while structured sampling also remains relatively stable. In contrast, Poisson sampling is more sensitive to γ , particularly at smaller strides. We next explain these observations by analyzing how the stride affects the number of w -span segments that can be sampled within a batch. Let $|R|$ denote the length of the time series, ℓ the segment length, and w the length of the protected scope of the time series. Ignoring boundary effects and integer rounding, $|S| \approx \frac{|R|-\ell}{\gamma} + 1$, $n_s \approx \frac{w+\ell-1}{\gamma}$. Thus, the fraction of w -span segments is approximately $q = \frac{n_s}{|S|} \approx \frac{w+\ell-1}{|R|-\ell+\gamma}$. Since the time series length is typically much larger than the stride, i.e., $|R| - \ell \gg \gamma$, this can be further approximated as $q \approx \frac{w+\ell-1}{|R|-\ell}$, indicating that the fraction of w -span segments is largely insensitive to γ . Structured sampling constructs each batch of size K via sampling with replacement [38]. Since each draw independently selects a segment from the entire segment set, the number of sampled w -span segments follows a binomial distribution, with each draw

having probability $n_s/|S|$ of selecting a w -span segment. Since this fraction changes little with γ , the resulting privacy accounting and utility are also relatively stable across different strides. Poisson spaced sampling exhibits a similar behavior, as its partitioning mechanism limits the number of w -span segments that can enter the same batch. In contrast, Poisson sampling independently includes each segment with a sampling probability determined by the batch size K relative to the total number of segments. Accordingly, the number of sampled w -span segments follows a binomial distribution over the n_s w -span segments. Although its expected value remains nearly unchanged as γ varies, decreasing γ increases n_s and allows outcomes in which more w -span segments are sampled simultaneously. In RDP accounting, an outcome involving j sampled w -span segments contributes through an exponential term of the form $\exp((\alpha - 1)\epsilon_\alpha(j))$, where $\epsilon_\alpha(j)$ increases with j . Thus, rare outcomes with large j can contribute disproportionately to the per-step privacy cost. Since RDP composes additively across training steps, i.e., $\epsilon_\alpha^{\text{total}} = T\epsilon_\alpha^{\text{step}}$, a larger per-step privacy cost reduces the number of allowable training steps under a fixed privacy budget.

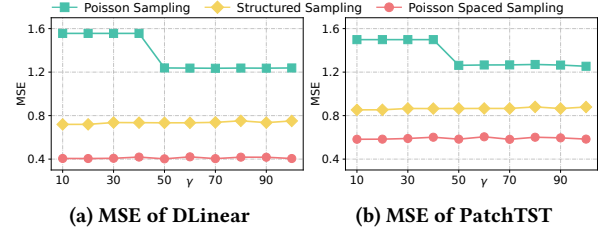


Figure 7: Results on the LargeST Benchmark Dataset with w set to 0.5% of the time series length, the sequence length $\ell = 100$, the batch size $K = 50$, and the noise multiplier $\sigma = 5$ with varying stride γ .

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